

A Calibration Method for On-Vehicle AR-HUD System Using Mixed Reality Glasses

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ABSTRACT

Calibration is a key step for on-vehicle AR-HUD systems to ensure the augmented information to be correctly viewed by the driver. State-of-art calibration methods require setting up of spatial tracking devices or attaching markers on vehicles, which is time-consuming and error-prone. In this paper, we present a novel multi-viewpoints calibration method for AR-HUD using only a mixed reality glasses such as HoloLens. The full calibration process can be done in one minute and provides high precise calibration result, while no markers need to be attached on vehicle.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—Interaction devices—Displays and imagers

1 INTRODUCTION

Head-up display (HUD) is a transparent display that can present information in front of users' eyes so that users can see the information without moving sight away. AR-HUD is a special kind of HUD which can provide driving related information directly in the driving environment through windshield, such as overlaying a navigation path on the road, marking pedestrians with surrounding rectangles, or sticking labels on interesting buildings. It is challenge to keep virtual image registered accurately with real world objects with on-vehicle AR-HUD due to the complex optical paths introduced by freeform reflectors in the system, such as the windshield.

There are some researches on the calibration for AR-HUD. Wu et al. [4] built an indoor system prototype to simulate the actual driving situation on the road. They mainly focused on the distortion correction with fixed viewing angle and fixed head position. They projected a pattern and built a function to avoid the windshield distortion based on pre-warped points and interpolation. It provides primitive idea of augmenting the reality by 2D overlaying but lacking 3D registration of virtual world in the real world, which makes it unpractical for real car AR applications. Wientapper et al. [3] decomposed the calibration into two phases: view-independent geometry of the virtual plane and view-dependent image warping which is similar to the display-relative calibration method proposed by Owen et al [1]. For correcting the distortion, they employed a higher-order polynomial function of 5 parameters. They used vision-based tracking method by attaching textured patterns on the windshield to achieve accurate camera registration with the help of Structure-from-Motion techniques. The setup was cheap but time-consuming during the preparation. Kaho Ueno and Takashi Komuro [2] took the diversity of road surface and practical face tracking into consideration. They used a depth camera to measure the road surface in

real time and a camera to perform face tracking to reduce the effect of head movement.

This paper proposes a novel calibration method for AR-HUD which is robust, convenient and fast. Our calibration model treats the optical system of AR-HUD as a black box, thus is more applicable. Furthermore, a mixed-reality glasses is used to simplify the calibration process. With the calibration result, a GPU-based real-time undistortion process is run on AR-HUD to provide correct virtual image to driver.

2 CALIBRATION MODEL

Our calibration model regards the complex optical system as a black-box. To overlay AR content in the driver's view, the virtual scene should be rendered from the viewpoint so that rays to the viewpoint in virtual scene coincides with the rays in real scene. Taking the viewpoint v fixed, the problem to solve is to get the direction r of the ray produced by each pixel p^s on the display. Mathematically, this can be represented as a mapping:

$$r_{ij} = f^v(p_{ij}^s) \quad (1)$$

To get the mapping, a camera is put at the viewpoint, capturing a predefined pattern displayed on the screen and getting the corresponding positions p_{ij}^c in camera image of p_{ij}^s on screen. Then r_{ij} can be computed by

$$r_{ij} = \mathbf{T}_c \mathbf{P}_c^{-1} p_{ij}^c \quad (2)$$

$\mathbf{T}_c$ is the camera's transform matrix to the vehicle space, and $\mathbf{P}_c$ is the camera's projection matrix. In this work, a grid of white blobs with black background is used as the pattern and an auto-detecting method is designed to extract corresponding position in camera image of each blob.

Like [3], we use polynomial functions as the multiple-viewpoint model. For each component of $\hat{r}_{ij}$ in single-viewpoint model, we have:

$$\hat{r}_{ij}^k = \sum_{d_1, d_2, d_3} b_{ijk}(d_1, d_2, d_3) x_v^{d_1} y_v^{d_2} z_v^{d_3}, d_i \in \mathbb{N}, \sum_i d_i \leq D \quad (3)$$

$k = 1, 2$ is the component index in $\hat{r}_{ij}$, (x_v, y_v, z_v) is the viewpoint position, $b_{ijk}(\cdot)$ are coefficients of the model, D is the maximal degree of the function. To train the model, hundreds of samples at different viewpoint in the feasible range should be taken, which form a training data set. Then, nonlinear regression method is used to calculate $b_{ijk}(\cdot)$.

3 CALIBRATION PROCESS

In this work, we use Microsoft HoloLens¹ as the calibration tool. We implement our calibration process with Unity3D². The first step is to calibrate the HoloLens' coordinate system with the vehicle coordinate system. We use ICP algorithm to calculate the transformation between HoloLens spatial mapping and our reference model.

¹<https://www.microsoft.com/en-us/hololens>

²<https://unity3d.com>

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Next step is taking mappings by HoloLens' front camera at different viewpoints to form the training data set. Normally 500 samples that approximately cover the feasible range are enough. Finally we train the model using Matlab³ nonlinear regression toolbox.

4 RESULTS

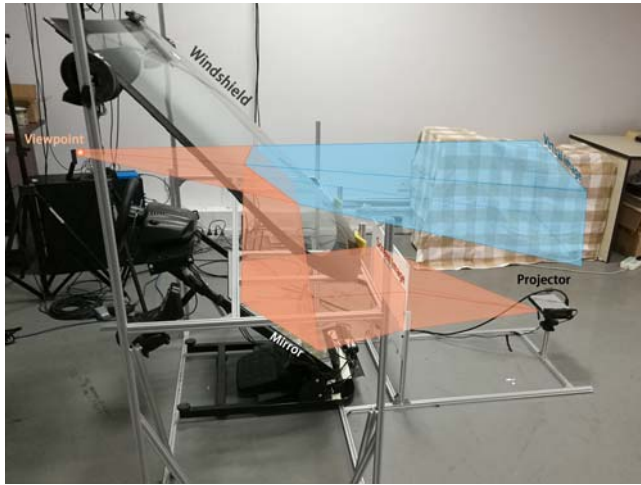


Figure 1: The indoor test-bed we built and the optical path of the system. The yellow path is rays from the projector to the viewpoint, and the blue path is virtual rays to the virtual image.

We build an indoor test-bed containing a windshield and a simple HUD prototype (Fig. 1). The projector is connected to a desktop computer. The virtual image seen by the user is about 3 meters away from the user, and its field-of-view is about 9° in horizontal and 3° in vertical. A demo is implemented by Unity3D to present the effects after calibration using test scenes. After the frame is rendered using standard rendering pipeline, undistortion is processed on GPU as a post-rendering step.

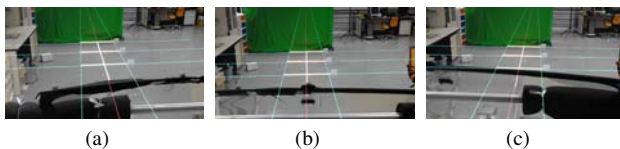


Figure 2: Camera tracking results. Four LED strips stuck on the floor is used as reference. the farthest strip is at distance of 6 meters.

To evaluate the camera tracking quality, we build a reference grid on the floor. A grid of same configuration is put in the virtual scene. If the user wearing HoloLens sees the virtual grid overlaying on the reference grid wherever he moves, the tracking can be considered reasonable. Fig. 2 shows the results captured from three different viewpoints. All results show precise matchings of the virtual grid and the reference grid.

To evaluate the undistortion result, we build two standard grid scenes: grid floor and grid tunnel. The grid floor scene contains horizontal and vertical lines in the $Y = 0$ plane, with distance up to 100 meters. The grid tunnel scene is 4 meters width and 2 meters high, with a grid wall at $Z = 30m$. Same Scenes are displayed in AR-HUD and HoloLens simultaneously. The position of HoloLens is sent to AR-HUD through UDP so that AR-HUD can render from

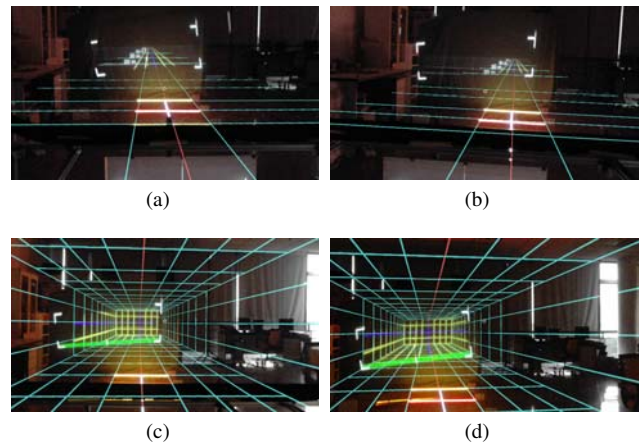


Figure 3: The undistorted result. Images in top row are the results of grid floor scene. Images in bottom row are the results of grid tunnel scene.

user's view and do correct undistortion. Fig. 3 shows the results of these two scenes at different viewpoints.

5 CONCLUSION

This paper proposes a calibration method for on-vehicle AR-HUD system using mixed-reality glasses. The optical model of AR-HUD is nonlinear, so we design a two-level mapping model to handle the distortion. The model is simple yet efficient at any viewpoint in feasible region. We implement the calibration process on HoloLens. Some demos are exhibited to verify the calibration result. Although our method is designed for on-vehicle AR-HUD system, it is interesting to investigate the capability of extension to other types of AR displays in future work.

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³<https://www.mathworks.com/products/matlab.html>